High-Order Omni-Attention Paradigm for Remote Sensing Image Fusion

Yuan Zhou, Wenyang Zhou, Xuanhua He, Xiaoran Shi, Xiaofeng Cong, Xiaopeng Han, Man Zhou, and Danfeng Hong

***Abstract*—Multimodal image fusion aims to comprehensively explore cross-modal information interactions between multiple source images, leveraging both spatial and channel dimensions. Existing methods primarily employ the cross-attention mechanism, which only exploits second-order properties in unidimensional modeling, thereby limiting higher order interactions and omni-attention-enhanced learning capabilities. To address this, we propose a high-order omni-attention paradigm for multimodal image fusion, adhering to the following principles: 1) high order. Current promising methods stack cascaded self-attention-driven transformers to achieve high-order capabilities, resulting in substantial computational costs. Inspired by the self-attention saturation issue and its intrinsic first-order function, we argue against the necessity of computing attention scores in every cascaded layer. Instead, we perform attention operations in the initial layer and share the calculated attention scores across subsequent layers via attention transformations. This motivates us to shorten the resource-expensive cascaded attention chain within an effective high-order attention paradigm; and 2) omni-dimension. Regarding omni-dimensional attention score calculation, we reformulate each pixel in 3-D space as token and perform dense interactions through sequential elementwise spatial multiplication in the Fourier domain mathematically equivalent to global interactions and rotation-based channelwise self-attention mechanism simultaneously, to facilitate the discernment of interdependencies between source images. Inheriting the above merits, the tailored high-order core is capable of yielding omni-dimensional information interactions across spatial and channel axes simultaneously, culminating in 3-D modeling. Comprehensive experiments across various benchmarks in multimodal image fusion tasks demonstrate that harnessing high-order interactions significantly enhances our model's ability to exploit multimodal synergies, leading to superior performance over the state-of-the-art alternatives.**

***Index Terms*—High-order omni-attention paradigm, infrared, pan-sharpening, visible image fusion.**

I. Introduction

MULTISENSOR imagery, as seen in applications, such as pan-sharpening in remote sensing and the fusion of infrared and visible images in natural scenes, effectively captures complementary and holistic scene information. This methodology transcends the limitations of single-sensor imaging by integrating salient data from diverse sensors, resulting in a composite image suitable for subsequent processing or decision-making tasks. Consequently, multimodal image fusion emerges as a significant and enduring challenge in image processing, with profound implications for downstream applications, including object detection [1], pedestrian reidentification [2], semantic segmentation [3], and image stitching [4].

A. Second Order

Recently, remarkable advancements in deep learning (DL) have sparked a revolution within the multimodal image fusion community, driven by the vision transformer (ViT), which leverages self-attention and cross-attention mechanisms characterized by second-order spatial interactions, deviating from the traditional dot products among key, query, and value components [5]. The representative SwinFusion [6] develops self-attention and cross-attention mechanisms to adeptly model and integrate dependencies within both intradomain and interdomain features. Similarly, PanFormer [7] designs a customized cross-modality transformer that incorporates two value terms from disparate modalities to effectively capture their collaborations. However, the existing methods primarily employ the cross-attention mechanism, which only exploits second-order properties in a unidimensional and cascaded manner, thereby limiting higher order interactions and omni-attention-enhanced learning capabilities. Furthermore, the cascaded second-order interaction paradigm only captures multiple second-order interactions and struggles to balance commendable performance with resource-intensive computations, posing challenges for practical applications, as illustrated in Fig. 1. To address these challenges, our investigation reveals that self-attention fundamentally operates as a first-order linear weight function. This insight enables us to replace the conventional cascaded second-order interaction sequence with efficient high-order modeling through tailored attention sharing.

B. Omni-Attention

The above second-order interaction, implemented in either unidimensional modeling or through homogeneous aggregation schemes, limits its effective receptive field from including more comprehensive interactions across all axes, as illustrated in Fig. 2. Contemporary self-attention mechanisms generally achieve interaction between pixels by calculating the

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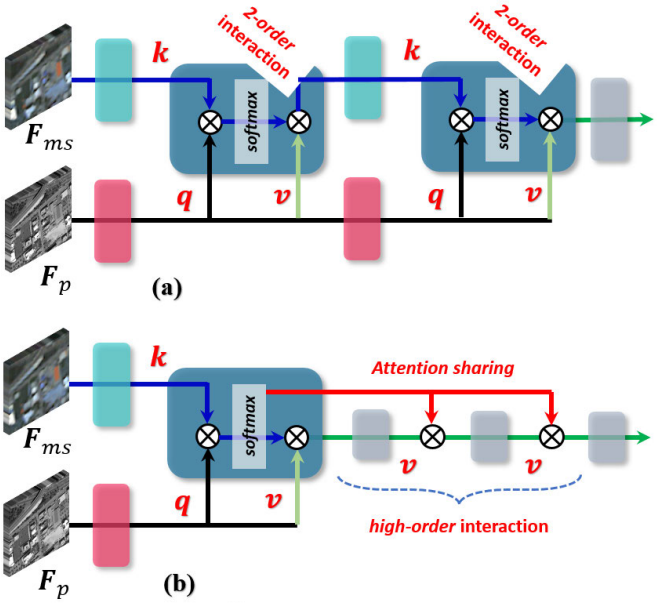


Fig. 1. Comparison between previous two-order cascaded chain and our proposed high-order paradigm. (a) Second-order interaction. (b) High-order interaction.

cross-covariance in either the spatial or channel direction, thereby exchanging context information in a unidimensional manner. This confinement of unidimensional operations to a fixed spatial range or solely within the channel dimension restricts the full potential of cross-modality information aggregation capability in dense 3-D-pixel interaction. Homogeneous aggregation schemes: These schemes perform simple hierarchical stacking of spatial attention and channel attention by decomposing the 3-D space into a chain of 2-D (H/W) space and 1-D (C) space. While the initial goal is to explore synergistic correlations between omni-dimensional information interactions across the spatial and channel axes simultaneously, culminating in comprehensive 3-D modeling, the stacking of homogeneous operators proves to be inefficient and suffers from premature saturation of the interaction range, resulting in a suboptimal effective receptive field. Within this endeavor, our focus is directed toward tailoring an effective and comprehensive omni-attention high-order paradigm that circumvents these challenges for multimodal image fusion tasks.

C. Solutions

In this work, we propose a high-order omni-attention paradigm for multimodal image fusion in Fig. 4, based on the following principles: 1) high order. Existing methods often rely on stacking cascaded self-attention-driven transformers to achieve high-order capabilities, which incurs significant computational costs. Addressing the challenge of self-attention saturation, we propose avoiding the computation of attention scores in every cascaded layer. Instead, attention operations are performed in the initial layer, with the resulting attention scores shared across subsequent layers via attention transformations. This approach reduces the resource-intensive cascaded attention chain while maintaining an effective high-order attention framework; and 2) omni-dimension. For omni-dimensional attention score calculation, we conceptualize each pixel in 3-D space as a token. We then perform dense interactions through sequential element-wise spatial multiplication in the Fourier domain, which is mathematically equivalent to global interactions, and apply a rotation-based channelwise self-attention mechanism simultaneously. This dual approach facilitates the identification of interdependencies between source images. Leveraging these principles, our high-order core effectively enables omni-dimensional information interactions across both spatial and channel axes, culminating in comprehensive 3-D modeling. Extensive experiments across various benchmarks in three key multimodal image fusion tasks—pan-sharpening, infrared, and visible image fusion—demonstrate that our model, by harnessing high-order interactions, significantly enhances the ability to exploit multimodal synergies, leading to superior performance compared to the state-of-the-art methods.

D. Evidence

Fig. 3 illustrates the attention maps across different layers of a cascaded ViT architecture, highlighting the effects of self-attention and cross-attention mechanisms on pan-sharpening. As observed, the attention maps from the self-attention mechanism, such as $\mathbf{A}_3$, $\mathbf{A}_4$, and $\mathbf{A}_5$, exhibit considerable similarity, reflecting a high degree of redundancy in feature representation as the network deepens. This redundancy aligns with the self-attention saturation problem, where attention matrices remain largely unchanged across layers. This phenomenon indicates that both self-attention and cross-attention mechanisms contribute to feature representation redundancy and unnecessary computational overhead. Given the current trend of employing cascaded self-attention-driven transformers to achieve high-order capabilities—often resulting in substantial computational costs—our findings suggest that it may not be essential to compute attention scores in every layer. Instead, by performing attention operations in the initial layer and sharing the computed scores across subsequent layers through attention transformations, we propose a more resource-efficient approach. As shown in Fig. 5, the illustration of the attention evolution mechanism at the bottom demonstrates how the tailored attention-sharing strategy at the top is transformed into an evolution mechanism by leveraging the matrix associative property. This mathematically equivalent transformation effectively mitigates attention collapse, reducing the computational burden of extensive cascaded attention chains while preserving high-order attention capabilities. The thorough investigation of this phenomenon, combining theoretical analysis with empirical observations, fully supports our claim.

In summary, the contributions of this work are as follows.

1) We propose a novel high-order omni-attention paradigm for multimodal image fusion, which integrates attention operations across multiple layers efficiently by sharing computed attention scores, thus reducing computational costs associated with traditional cascaded attention.

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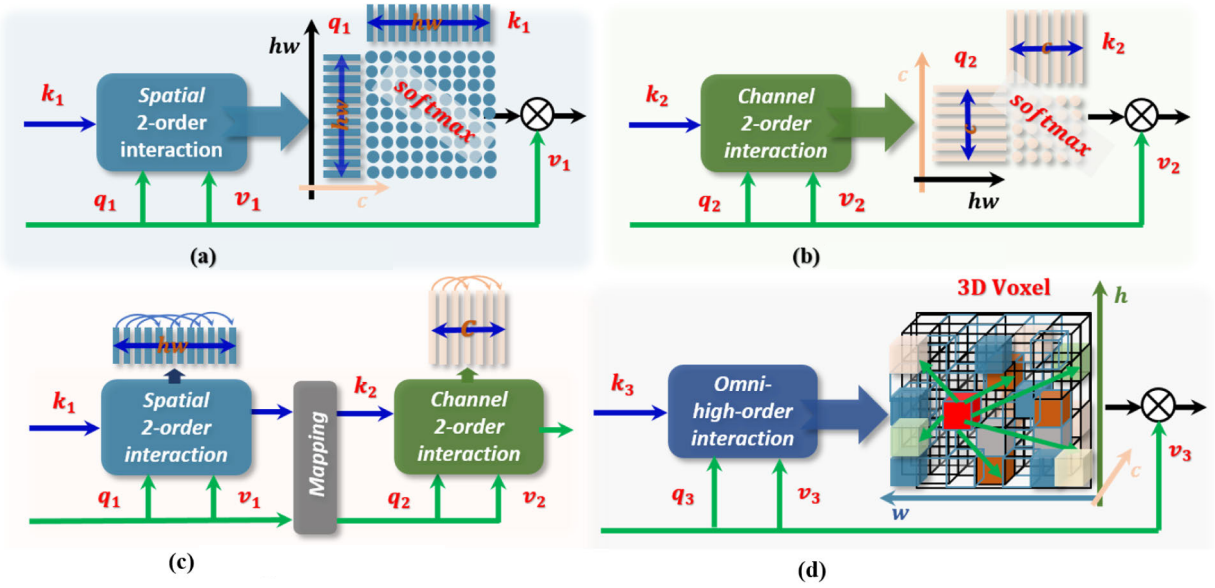


Fig. 2. Comparison between previous cross-attention and our proposed omni-attention paradigm in dense 3-D-pixel interaction. The typical (a) spatial attention and (b) channel attention mechanisms only model unidimensional interactions, focusing on spatial-only or channel-only relationships independently, and (c) homogeneous attention. In contrast, (d) our proposed omni-attention paradigm explores the synergistic correlations between omni-dimensional information interactions across the spatial and channel axes simultaneously, culminating in a comprehensive 3-D modeling.

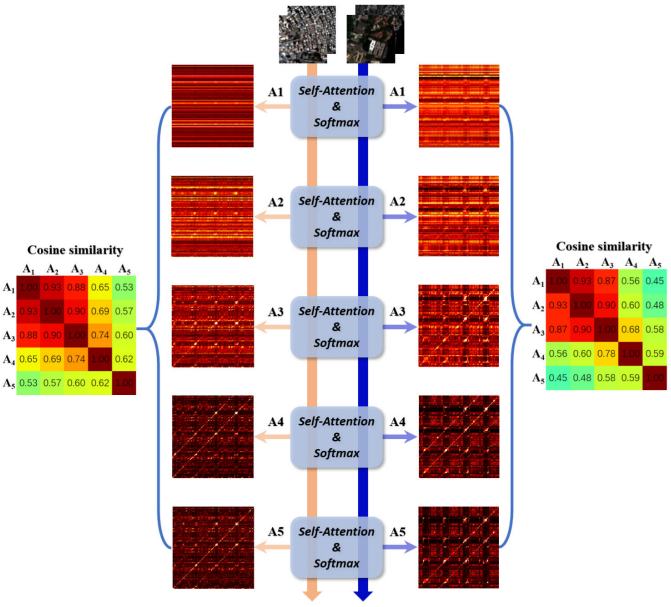


Fig. 3. Illustration of attention maps across different layers from a cascaded ViT architecture on pan-sharpening. $\mathbf{A}_i (i = 1, \ldots, 5)$ denotes the attention map from the ith ViT block.

2) We introduce a theoretically feasible approach for omni-dimensional attention score calculation by treating each pixel in 3-D space as a token, performing dense interactions through Fourier domain multiplication, and incorporating rotation-based channelwise self-attention mechanisms. This method enhances the model's ability to discern complex interdependencies between source images.
3) Extensive experiments across benchmark datasets for pan-sharpening, infrared, and visible image fusion validate the efficacy of our proposed paradigm. The high-order interactions greatly enhance the model's ability to leverage multimodal synergies, resulting in superior performance compared to the existing state-of-the-art methods.

II. Related Work

A. Pan-Sharpening

Traditional pan-sharpening methods are typically classified into three primary categories: component substitution (CS), multiresolution analysis (MRA), and variational optimization (VO) approaches. CS methods, such as intensity hue-saturation (IHS) fusion, principal component analysis (PCA), Brovey transforms, and Gram–Schmidt (GS) orthogonalization, are widely used [8], [9]. Enhancements to these methods include nonlinear IHS (NIHS) to reduce spectrum distortion and adaptive techniques like the GSA method [10], [11]. Despite their practicality, CS and MRA methods often introduce artifacts into the fused images. VO methods have emerged as alternatives to address spectral distortion and improve the spatial resolution of multispectral images. For example, P+XS pan-sharpening posits that the PAN image can be modeled as a linear combination of high-resolution multispectral (HRMS) bands, with the upsampled low-resolution multispectral (LRMS) image approximating a blurred HRMS image [12]. VO approaches incorporate constraints, such as dynamic gradient sparsity (SIRF), local gradient constraints (LGCs), and group low-rank constraints for texture similarity (ADMM) [13], [14], [15]. Despite their sophistication, VO methods often require manual parameter tuning and may struggle to capture structural relationships within images, potentially leading to suboptimal performance.

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Convolutional neural networks (CNNs) have transformed computer vision with their prowess in nonlinear fitting and feature extraction, making them crucial for hyperspectral and remote sensing image analysis [16]. Recent advancements in pan-sharpening focus on CNN-based approaches [17], [18]. Masi et al. [19] were among the first to apply CNNs to pan-sharpening, demonstrating superior results compared to traditional methods. Yang et al. [20] extended this work by incorporating residual blocks [21] into a deeper CNN architecture, while Yuan et al. [22] introduced a multiscale module to enhance the CNN structure. Cai and Huang [23] further improved performance by utilizing multiscale image inputs within the backbone network. A new category of model-driven CNNs has recently gained traction, integrating physical insights into optimization-based tasks. Xu et al. [24] applied distinct priors for PAN and MS images within a structured CNN framework, enhancing interpretability. Xie et al. [25] incorporated an optimization algorithm into a CNN architecture, while Tian et al. [26] and Wu et al. [27] combined VO techniques with deep residual CNNs. Zhou et al. [28] introduced a novel pan-sharpening framework driven by mutual information (MI), which enhances information representation through complementary learning between PAN and MS modalities, thereby reducing redundancy and significantly improving pan-sharpening performance.

B. Infrared and Visible Image Fusion

Infrared and visible image fusion aims to create a composite image that highlights salient features from the infrared spectrum while preserving fine textures from the visible spectrum. Traditional methods often involve transforming input images into specific domains through mathematical operations and then applying fusion strategies. Key approaches include multiscale transforms (MSTs), sparse representation (SR) [29], fuzzy set-based methods [30], and hybrid techniques [31]. MST-based methods, a prominent category, typically follow a three-step process [32]: 1) extracting scale-specific coefficients from source images using transformation techniques; 2) merging these coefficients through predefined fusion strategies; and 3) reconstructing the fused image via inverse transformation. The effectiveness of these methods relies on the chosen transformation techniques and fusion rules, which significantly impact fusion quality. Common decomposition techniques include curvelet [33], contourlet [34], wavelet [35], Laplacian pyramid [36], and shearlet transforms [37]. Basic fusion methods, such as maximum selection, simple averaging, and weighted averaging, are often used but may introduce checkerboard artifacts. Although more advanced techniques can enhance fusion quality, they often come with increased computational complexity and processing time.

Recent advancements in DL have addressed the limitations of conventional methods. DL-based fusion algorithms integrate feature extraction, fusion, and image reconstruction into a unified model framework [38]. Liu et al. [39] first applied CNNs to image fusion by computing binary maps for multifocus images. Li and Wu [40] further developed an encoder–decoder model, using the encoder for feature preservation and the decoder for fusion result reconstruction. This was enhanced by Li et al. [41] with a residual fusion network, RFN-Nest, incorporating a two-stage training strategy. Ma et al. [42] introduced FusionGAN, a generative adversarial network (GAN) approach, which models the fusion process as a min–max game between a generator and a discriminator. They advanced this with a dual-discriminator conditional GAN (DDcGAN) for fusing the images of different resolutions [43]. Yang et al. [44] presented a texture conditional GAN (TC-GAN) and Ma et al. [45] extended FusionGAN to handle multidistribution simultaneous estimation. Xu et al. [46] proposed a unified fusion framework with continuous learning to address catastrophic forgetting, and Liu et al. [47] focused on lightweight model design and adaptive training strategies for image fusion and semantic perception tasks.

III. Proposed Method

In this section, we first present the comprehensive flowchart of our approach, as illustrated in Fig. 4. We then explore the motivation behind our design, focusing on the intrinsic modeling mechanism of self-attention as a first-order weighting function and its theoretically justified and effective formulation in the Fourier domain. Following this, we provide an in-depth discussion of our core high-order omni-attention modeling. Notably, we illustrate the application of our method using pan-sharpening, with infrared and visible image fusion and medical image fusion serving as additional relevant examples.

A. Overview Framework

Given a PAN image, $I_{\mathcal{P}} \in \mathbb{R}^{H\times W\times 1}$, and an LRMS image, $I_{\mathcal{M}} \in \mathbb{R}^{h\times w\times C}$, we adopt the separate dual-branch modality-aware encoders to project $I_{\mathcal{P}}$ and the upsampled $I_{\mathcal{M}}$, yielding $F_{\mathcal{P}} \in \mathbb{R}^{H\times W\times C}$ and $F_{\mathcal{M}} \in \mathbb{R}^{H\times W\times C}$. Within each branch, we employ the half-instance normalization module as the basic unit. Subsequently, the extracted modality-aware shallow-level features are passed through the proposed core omni-attention high-order modeling paradigm in a sequential manner as follows:

$$F_{\mathcal{P}}, F_{\mathcal{M}} = \mathrm{E}_{\mathcal{P}}\left(I_{\mathcal{P}}\right), \mathrm{E}_{\mathcal{M}}\left(I_{\mathcal{M}}\right) \tag{1}$$

where $\mathrm{E}_{\mathcal{P}}(\cdot)$ and $\mathrm{E}_{\mathcal{M}}(\cdot)$ signify the feature extraction encoders for the PAN and multispectral modalities, respectively. Then, we employ the successively designed omni-attention high-order modeling paradigm, yielding omni-dimensional information interactions across spatial and channel axes simultaneously, culminating in a 3-D modeling capability across modality-aware features $F_{\mathcal{P}}$ and $F_{\mathcal{M}}$

$$F_{\mathcal{H}} = \mathrm{HOAB}_{L}\left(F_{\mathcal{P}}, F_{\mathcal{M}}\right) \tag{2}$$

where L indicates the iteration number of our high-order omni-attention block (HOAB). Finally, the transformed deep-level features are projected back into the image space to generate the fused result, $I_{\mathcal{F}} \in \mathbb{R}^{H\times W\times C}$ using the core HINet block from the encoder in conjunction with the 1×1 convolution unit as follows:

$$I_{\mathcal{F}} = \mathrm{D}_{\mathcal{C}}\left(F_{\mathcal{H}}\right) + \mathtt{Up}\left(I_{\mathcal{M}}\right) \tag{3}$$

where $\mathtt{Up}(.)$ and $\mathrm{D}_{\mathcal{C}}(\cdot)$ represent the upsampling operator and the corresponding decoder, respectively.

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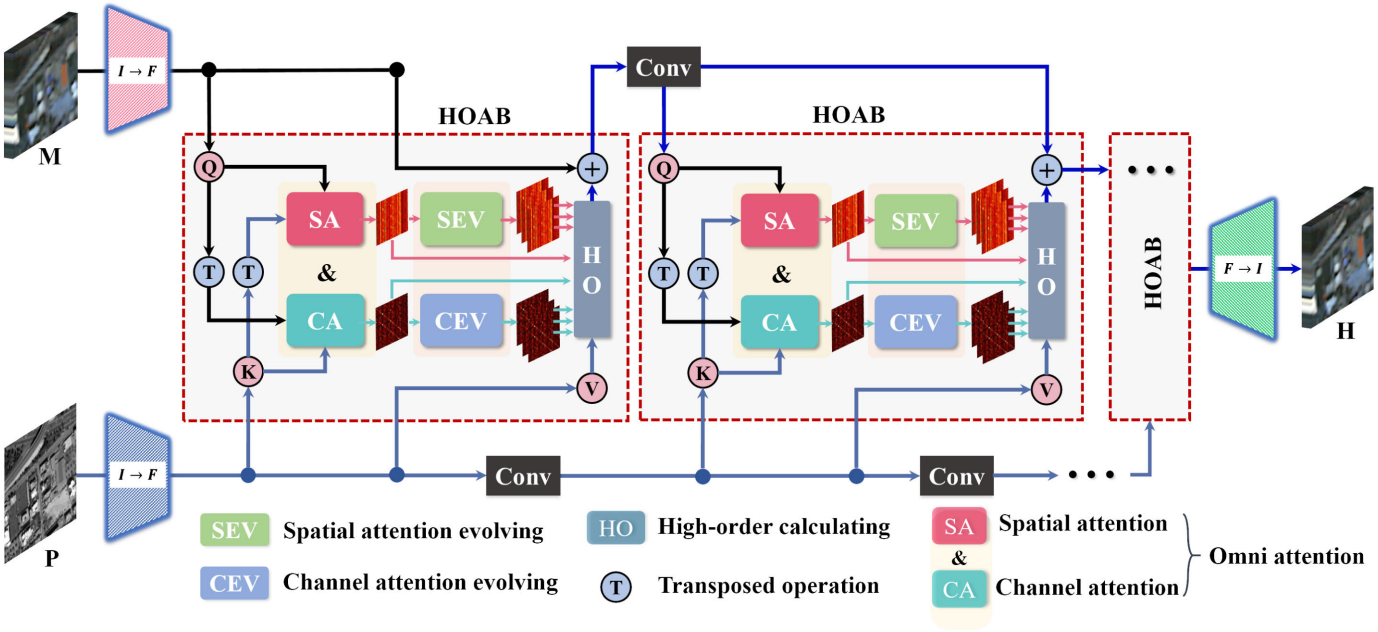


Fig. 4. Detailed framework of the proposed omni-attention HOAB. HOAB introduces a comprehensive approach to attention score calculation by treating each pixel in 3-D space as an individual token. This framework employs a combination of dense interactions through Fourier domain multiplication-driven spatial attention and rotation-based channelwise self-attention mechanisms and shares computed attention scores from the initial layer across subsequent layers via attention transformations, thus reducing computational costs.

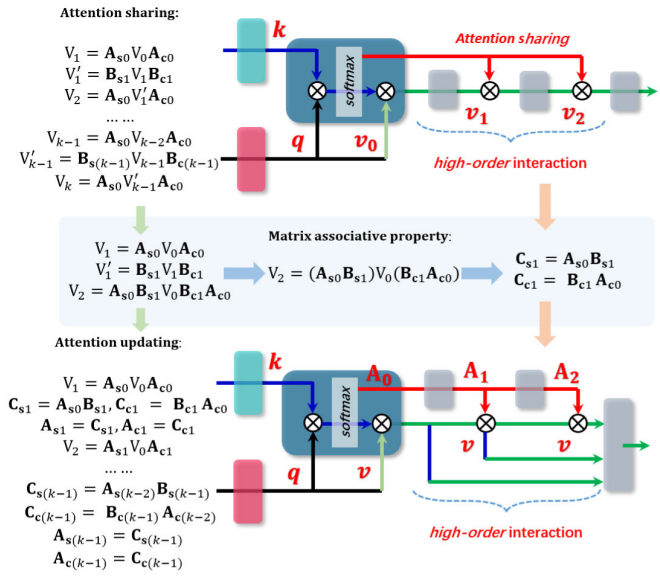


Fig. 5. Illustration of the attention evolution mechanism at the bottom involves transforming the tailored attention-sharing strategy shown at the top into an evolution mechanism by leveraging the matrix associative property. This mathematically equivalent transformation is capable of effectively further mitigating the issue of attention collapse.

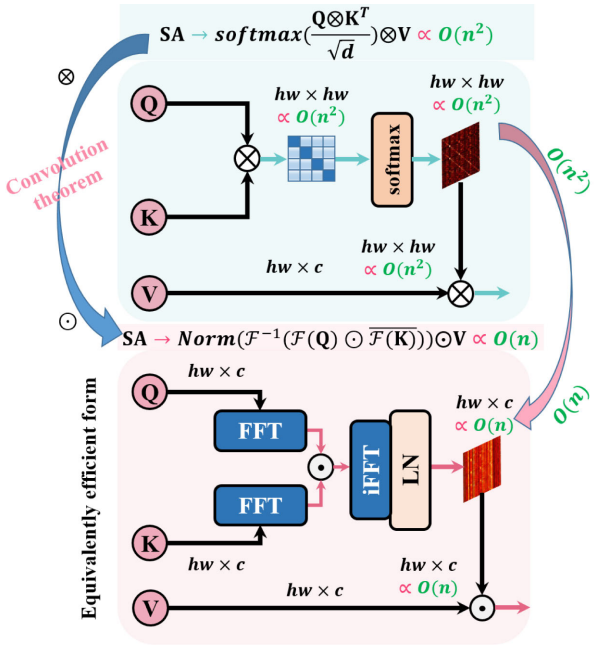


Fig. 6. Equivalently efficient form. The convolution theorem establishes that the correlation or convolution of two signals in the spatial domain equals the Hadamard product between them in the frequency domain.

B. Preliminary

1) High Order: The cross-attention mechanism plays a pivotal role in pan-sharpening by enabling effective fusion of complementary information from PAN and MS images. Specifically, texture-rich details from the PAN image are transferred to enhance the spectral representation of the MS image through second-order spatial interactions. In this framework, the PAN image serves as the key and value, while the MS image acts as the query, formulated as follows:

$$O_s^{(2)}\left(F_p\right) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \tag{4}$$

While effective for pairwise relationships, this approach fails to capture the intricate dependencies that emerge from simultaneous multifeature interactions. Our higher order formulation

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extends this to Nth-order interactions through functional composition

$$F^{(n)} = F_{(n)} \circ F_{(n-1)} \circ \cdots \circ F_{(2)} \circ F_{(1)} \tag{5}$$

where each $F_{(i)}$ represents a second-order transformation layer, and the composition captures complex hierarchical relationships.

C. Spatial Interactions of $\mathbf{D}_3 = \mathbf{S}_2^{Hw} \circ \mathbf{C}_1^C$

1) Cross-Modal Attention for Pan-Sharpening: The cross-attention mechanism plays a fundamental role in pan-sharpening by enabling effective fusion of complementary information from PAN and MS images. This mechanism facilitates global modeling of cross-modal correlations through second-order spatial interactions among the key, query, and value components. Specifically, texture-rich information from the PAN image is transferred to enhance the spectral representation of the MS image. In this fusion framework, the PAN image serves as the key and value, while the MS image acts as the query, formulated as follows:

$$\mathbf{Q} = F_{\mathcal{M}}\mathbf{W}^Q, \quad \mathbf{K} = F_{\mathcal{P}}\mathbf{W}^K, \quad \mathbf{V} = F_{\mathcal{P}}\mathbf{W}^V \tag{6}$$

where $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$ denote linear transformations for modality-specific feature projection in the fusion process.

The cross-attention mechanism for pan-sharpening, capturing second-order spatial interactions between PAN and MS modalities, is expressed as follows:

$$\mathcal{O}_S\left((F_{\mathcal{P}})^2\right) = F^1_{\mathcal{P}_S} = \text{softmax}\left(\frac{\mathbf{Q}\otimes\mathbf{K}^T}{\sqrt{d_k}}\right)\otimes\mathbf{V} = \mathbf{A}_s\otimes\mathbf{V} \tag{7}$$

where d_k represents the dimension of the key, $\otimes$ indicates the dot-product operation, $\mathbf{A}_s \in \mathbb{R}^{HW\times HW}$ is the cross-modal correlation matrix, and $\mathcal{O}_S((F_{\mathcal{P}})^2)$ signifies the output of the cross-attention module for image fusion. Although achieving global spatial interactions beneficial for pan-sharpening, the matrix multiplication incurs significant computational costs due to quadratic complexity, limiting practical application for high-order fusion operations.

2) Frequency-Domain Fusion Formulation: Each element of the attention matrix $\mathbf{A}$ can be expressed as $\mathbf{A}_{sij} = \langle \mathbf{q}_i, \mathbf{k}_j\rangle$, where $\mathbf{q}_i \in \mathbf{Q}$ and $\mathbf{k}_j \in \mathbf{K}$, with $\langle\cdot\rangle$ denoting the inner product. According to the convolution theorem, the correlation operation in spatial domain is equivalent to Hadamard product in frequency domain, enabling efficient cross-modal fusion. This property allows us to reformulate the attention mechanism for computational efficiency in image fusion tasks.

The modality-aware features $F_{\mathcal{P}}$ and $F_{\mathcal{M}}$ are transformed into frequency domain using fast Fourier transform (FFT), with correlation computed as follows:

$$\mathbf{A}_s = \mathcal{F}^{-1}\left(\mathcal{F}\left(F_{\mathcal{M}}\mathbf{W}^Q\right)\odot\overline{\mathcal{F}\left(F_{\mathcal{P}}\mathbf{W}^K\right)}\right) \tag{8}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ represent the FFT and inverse FFT, $\odot$ denotes the Hadamard product, and $\overline{\mathcal{F}(\cdot)}$ indicates the conjugate transpose operation.

The fused features incorporating second-order spatial interactions for pan-sharpening are obtained as follows:

$$\mathcal{O}_S\left((F_{\mathcal{P}})^2\right) = F^1_{\mathcal{P}_S} = \text{Norm}\left(\mathbf{A}_s\right)\odot\left(F_{\mathcal{P}}\mathbf{W}^V\right) \tag{9}$$

where Norm(.) denotes the layer normalization.

3) Mathematical Foundation for Fusion Efficiency: The cross-attention computation in spatial domain with quadratic complexity is expressed as follows:

$$\mathbf{A}_s = \text{softmax}\left(\frac{\mathbf{Q}\otimes\mathbf{K}^T}{\sqrt{d_k}}\right). \tag{10}$$

Based on spectrum convolution theory, we derive an efficient in Fig. 6, we derive an efficient frequency-domain implementation for fusion tasks

$$\mathbf{A}_s = \mathcal{F}^{-1}\left(\mathbf{Q}\otimes\overline{\mathcal{F}(\mathbf{K})}\right). \tag{11}$$

Examining each tensor vector of $\mathbf{Q}\otimes\mathbf{K}^T$, it can be reformulated as follows:

$$\begin{aligned}\mathbf{A}_{sij} &= \left\langle \mathbf{q}_i, \mathbf{k}_j^T\right\rangle = \mathbf{q}_{i1}\cdot\mathbf{k}_{1j} + \mathbf{q}_{i2}\cdot\mathbf{k}_{2j} + \cdots + \mathbf{q}_{ihw}\cdot\mathbf{k}_{hwj} &(12)\\ &= \mathbf{q}_{i1}\cdot\left(\mathbf{k}_j^T\right)_1 + \mathbf{q}_{i2}\cdot\left(\mathbf{k}_j^T\right)_2 + \cdots + \mathbf{q}_{ihw}\cdot\left(\mathbf{k}_j^T\right)_{hw}\\ &= \sum_{i,j}\mathbf{q}_i\mathbf{k}_{hw-j} = \mathbf{q}_i * \mathbf{k}_j &(13)\end{aligned}$$

where $*$ denotes the correlation-convolution calculation, demonstrating efficient attention computation in Fourier domain for image fusion.

The mathematical proof establishes the frequency-domain equivalence

$$\begin{aligned}\mathcal{F}\left(\mathbf{q}_i * \mathbf{k}_j\right) &= \int_a^b\int_a^b \mathbf{q}_i(t)\,\mathbf{k}_j(t-\tau)\,d_\tau\mathbf{e}^{-zwt}d_t &(14)\\ &= \int_a^b \mathbf{q}_i(t)\int_a^b \mathbf{k}_j(t-\tau)\,\mathbf{e}^{-zwt}d_t d_\tau &(15)\end{aligned}$$

where $z^2 = -1$. The Fourier transformation derivation continues as follows:

$$\mathcal{F}\left(\mathbf{k}_j(t-\tau)\right) = \int_a^b \mathbf{k}_j(t-\tau)\,\mathbf{e}^{-zwt}d_t. \tag{16}$$

With variable substitution $t-\tau = r$

$$\mathcal{F}\left(\mathbf{k}_j(t-\tau)\right) = \mathcal{F}\left(\mathbf{k}_j(r) = \int_a^b \mathbf{k}_j(r)\,\mathbf{e}^{-zw(r+\tau)}d_r\right) \tag{17}$$

$$= \int_a^b \mathbf{k}_j(r)\,\mathbf{e}^{-zw(r)}d_r\mathbf{e}^{-zw\tau} = \mathcal{F}\left(\mathbf{k}_j\right)\mathbf{e}^{-zw\tau}. \tag{18}$$

Using $\mathcal{F}(\mathbf{k}_j(t-\tau)) = \mathcal{F}(\mathbf{k}_j)\mathbf{e}^{-zw\tau}$, we derive

$$\begin{aligned}\mathcal{F}\left(\mathbf{q}_i * \mathbf{k}_j\right) &= \int_a^b \mathbf{q}_i(t)\int_a^b \mathbf{k}_j(t-\tau)\,\mathbf{e}^{-zwt}d_t d_\tau &(19)\\ &= \int_a^b \mathbf{q}_i(t)\,\mathcal{F}\left(\mathbf{k}_j\right)\mathbf{e}^{-zw\tau}d_\tau &(20)\\ &= \mathcal{F}\left(\mathbf{k}_j\right)\int_a^b \mathbf{q}_i(t)\,\mathbf{e}^{-zw\tau}d_\tau &(21)\\ &= \mathcal{F}(\mathbf{q}_i)\,\mathcal{F}\left(\mathbf{k}_j\right). &(22)\end{aligned}$$

Final equivalence is established as follows:

$$\mathbf{q}_i * \mathbf{k}_j = \mathcal{F}^{-1}\left(\mathcal{F}\left(\mathbf{q}_i * \mathbf{k}_j\right)\right) = \mathcal{F}^{-1}\left(\mathcal{F}(\mathbf{q}_i)\,\mathcal{F}\left(\mathbf{k}_j\right)\right). \tag{23}$$

The normalized attention map Norm($\mathbf{A_s}$) represents cross-modal correlations in fusion. Layer normalization transforms input distribution to approximate Gaussian distribution through

$$\mu = \frac{1}{m}\sum_{i=1}^m x_i, \quad \sigma = \sqrt{\frac{1}{m}\sum\nolimits_{i=1}^m (x_i-\mu)^2} \tag{24}$$

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$$\hat{x}_i = \frac{x_i - \mu}{\sigma}. \tag{25}$$

Building on distributional properties, high-order spatial modeling for fusion becomes feasible through Fourier-based formulation and channelwise mixing

$$\begin{aligned} F_{\mathcal{P}} &\to \mathcal{O}_{\mathbf{S}}\left((F_{\mathcal{P}})^N\right) \to F^1_{\mathcal{P}_\mathcal{S}} \to \mathcal{O}_{\mathbf{C}}\left(\left(F^1_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^1_{\mathcal{P}_\mathcal{C}} \\ &\ldots \to \mathcal{O}_{\mathbf{S}}\left(\left(F^i_{\mathcal{P}_\mathcal{C}}\right)^N\right) \to F^{i+1}_{\mathcal{P}_\mathcal{S}} \ldots \mathcal{O}_{\mathbf{C}}\left(\left(F^{L-1}_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^L_{\mathcal{P}_\mathcal{C}}. \end{aligned} \tag{26}$$

This process captures second-order interactions but suffers from computational complexity. The decomposition $\mathbf{D}_3 \approx \mathbf{S}_2^{HW} \circ \mathbf{C}_1^C$ is expressed as follows:

$$\mathbf{S}_2^{HW}: \ \mathbf{v}_{1b} = \text{softmax}\left(\mathbf{k}_1, \mathbf{q}_1\right) \cdot \mathbf{v}_1 \tag{27}$$

$$\mathbf{v}_2 = \texttt{Mapping}\left(\mathbf{v}_{1b}\right) \neq \mathbf{v}_1 \tag{28}$$

$$\mathbf{C}_1^C: \ \mathbf{O} = \text{softmax}\left(\mathbf{k}_2^T, \mathbf{q}_2^T\right) \cdot \mathbf{v}_2 \tag{29}$$

where Mapping(.) denotes the multilayer perceptron.

Our omni-attention high-order fusion paradigm improves cross-modal interaction

$$\begin{aligned} F_{\mathcal{P}} &\to \mathcal{O}_{\mathbf{SC}}\left((F_{\mathcal{P}})^N\right) \to F^1_{\mathcal{P}_\mathcal{S}} \to \mathcal{O}_{\mathbf{SC}}\left(\left(F^1_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^2_{\mathcal{P}_\mathcal{S}} \\ &\ldots \to \mathcal{O}_{\mathbf{SC}}\left(\left(F^i_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^{i+1}_{\mathcal{P}_\mathcal{S}} \ldots \mathcal{O}_{\mathbf{SC}}\left(\left(F^{L-1}_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^L_{\mathcal{P}_\mathcal{S}}. \end{aligned} \tag{30}$$

The omni-attention mechanism $\mathcal{O}_{\mathbf{SC}}: \mathbf{D}_3 = \mathbf{S}_2^{HW} \circ \mathbf{C}_1^C$ is formulated as follows:

$$\mathbf{S}_2^{HW}: \mathbf{v}_{1b} = \text{softmax}\left(\mathbf{k}_1, \mathbf{q}_1\right) \cdot \mathbf{v}_1 \tag{31}$$

$$\mathbf{C}_1^C: \mathbf{O} = \text{softmax}\left(\mathbf{k}_1^T, \mathbf{q}_1^T\right) \cdot \mathbf{v}_{1b}. \tag{32}$$

Theoretical equivalence is established as $3-\text{D}-\text{pixel}(\mathbf{v}_1) = \text{softmax}(\mathbf{k}_1^T, \mathbf{q}_1^T) \otimes \texttt{softmax}(\mathbf{k}_1, \mathbf{q}_1) \cdot \mathbf{v}_1$, with practical implementation

$$\mathbf{S}_2^{HW}: \mathbf{v}_{1b} = \text{Norm}\left(\mathbf{A}_s\right) \cdot v_1 \tag{33}$$

$$\mathbf{C}_1^C: O = \text{softmax}\left(\mathbf{k}_1^T, \mathbf{q}_1^T\right) \otimes \mathbf{v}_{1b}. \tag{34}$$

For efficient fusion, we employ attention sharing across layers

$$\mathbf{A}_s^i = \texttt{Conv}_{1\times1}\left(\mathbf{A}_s^{i-1}\right) \tag{35}$$

$$\mathbf{S}_2^{HW}: \mathbf{A}_s^0 \leftarrow \mathcal{F}^{-1}\left(\mathbf{Q} \otimes \overline{\mathcal{F}\left(\mathbf{K}\right)}\right) \tag{36}$$

where $\mathbf{A}_s^{i-1}$ and $\mathbf{A}_s^i$ represent the adjacent attention calculations in the fusion pipeline.

D. Channel Interaction of $\mathbf{D}_3 = \mathbf{S}_2^{Hw} \circ \mathbf{C}_1^C$

Channelwise interactions are crucial for spectral quality in pan-sharpening, explicitly capturing interdependencies between feature channels. The channelwise cross-attention formulates dependencies between PAN and MS features

$$\mathbf{a}_{i,j} = \sum_{m=0}^{C}\sum_{n=0}^{C} \frac{\mathbf{q}_{i,m}\mathbf{k}_{n,j}}{\sum_{i=1}^{C}\mathbf{q}_{i,m}\mathbf{k}_{n,j}} \tag{37}$$

where $\mathbf{q}_{i,m} \in Q$, $\mathbf{k}_{n,j} \in K$, $\mathbf{a}_{i,j} \in A$, and feature projections are

$$\mathbf{Q} = F_{\mathcal{M}}\mathbf{W}^Q, \quad \mathbf{K} = F_{\mathcal{P}}\mathbf{W}^K, \quad \mathbf{V} = F_{\mathcal{P}}\mathbf{W}^V. \tag{38}$$

The output $\mathbf{O} \in \mathbb{R}^{H\times W\times C}$ is obtained as follows:

$$\mathbf{O}_{k,s} = \sum_{m=0}^{C}\sum_{n=0}^{C} \mathbf{a}_{l,m}\mathbf{v}_{n,s} = \mathcal{O}_1\left(\mathbf{V}\right) \tag{39}$$

where $\mathcal{O}_1(\cdot)$ indicates the first-order weights.

Analyzing the intrinsic first-order weights, with $\mathbf{k}^i$ and $\mathbf{q}^j$ as reshaped feature vectors

$$\mathbf{O}_j = \text{softmax}\left(\mathbf{q}_i, \mathbf{k}_j\right) \cdot \mathbf{v}_j \quad \forall i \tag{40}$$

$$0 < \text{softmax}\left(\mathbf{q}_i, \mathbf{k}_j\right) < 1, \ \sum_i \text{softmax}\left(\mathbf{q}_i, \mathbf{k}_j\right) = 1 \quad \forall i. \tag{41}$$

Detailed formulation

$$\text{softmax}\left(\mathbf{k}_i, \mathbf{q}_j\right) = \frac{\texttt{exp}\left((\mathbf{q}_i)^T\right)\mathbf{k}_j}{\sum_{l=1}^{C}\texttt{exp}\left((\mathbf{q}_i)^T\right)\mathbf{k}^l}. \tag{42}$$

This normalization transforms exponentials to probability distribution. Mathematically, for any function $p(x)$ satisfying $0 \le p(x) \le 1, \sum_x p(x) = 1$

$$p\left(\mathbf{v}_j\right) \propto \text{softmax}\left(\mathbf{q}_i, \mathbf{k}_j\right) \tag{43}$$

$$\mathbf{O}_j = \int_0^1 p\left(\mathbf{v}_j\right)\mathbf{v}_j d\mathbf{v} \approx \mathbf{E}\left(\mathbf{v}_j\right) \tag{44}$$

where $\mathbf{E}(.)$ signifies first-order expectation calculation. For probability distribution $p(\mathbf{v})$ satisfying constraints

$$0 \le p\left(\mathbf{v}\right) \le 1, \quad \sum_i p\left(\mathbf{v}\right) = 1. \tag{45}$$

Expectation is expressed as follows:

$$\mathbf{E}\left(\mathbf{v}_j\right) = \int_0^1 p\left(\mathbf{v}\right)\mathbf{v}_j d\mathbf{v}. \tag{46}$$

The softmax(.) function serves as probability sampling distribution, revealing that attention operates as first-order linear weight function implementable via 1-D convolution.

1) High-Order Channel Fusion: While spatial attention utilizes channel dimension, it does not transmit interchannel information. Our high-order omni-attention addresses this limitation for improved fusion

$$\mathbf{S}_2^{HW}: \mathbf{v}_{1b} = \text{Norm}\left(\mathbf{A}_c\right) \cdot \mathbf{v}_1 \tag{47}$$

$$\mathbf{C}_1^C: \mathbf{O} = \text{softmax}\left(\mathbf{k}_1^T, \mathbf{q}_1^T\right) \cdot \mathbf{v}_{1b}. \tag{48}$$

The multiorder interaction chain for progressive fusion refinement

$$\begin{aligned} F_{\mathcal{P}} &\to \mathcal{O}_{\mathbf{SC}}\left((F_{\mathcal{P}})^N\right) \to F^1_{\mathcal{P}_\mathcal{S}} \to \mathcal{O}_{\mathbf{SC}}\left(\left(F^1_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^2_{\mathcal{P}_\mathcal{S}} \\ &\ldots \to \mathcal{O}_{\mathbf{SC}}\left(\left(F^i_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^{i+1}_{\mathcal{P}_\mathcal{S}} \ldots \mathcal{O}_{\mathbf{SC}}\left(\left(F^{L-1}_{\mathcal{P}_\mathcal{S}}\right)^N\right) \to F^L_{\mathcal{P}_\mathcal{S}}. \end{aligned} \tag{49}$$

Linear updating strategy enables efficient high-order channel fusion

$$\mathbf{C}_1^C: \mathbf{A}_c^i = \texttt{Conv}_{1\times1}\left(\mathbf{A}_c^{i-1}\right) \tag{50}$$

where $\mathbf{A}_c^{i-1}$ and $\mathbf{A}_c^i$ represent the adjacent attention calculations in the channel fusion process.

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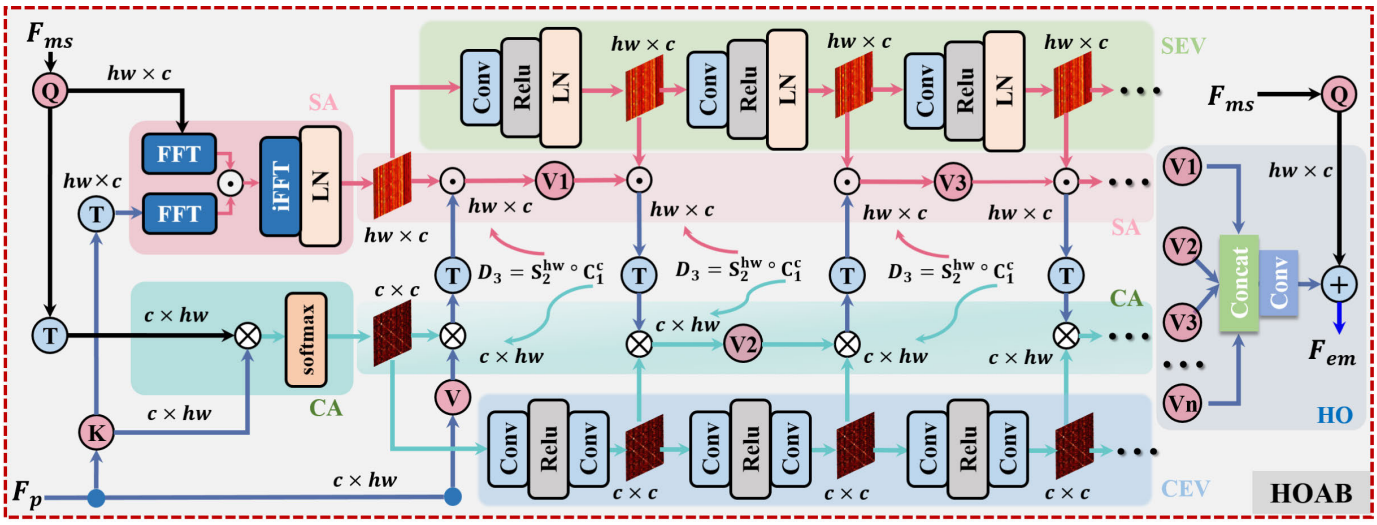


Fig. 7. Detailed framework of omni-dimensional high-order attention score calculation by treating each pixel in 3-D space as a token, performing dense interactions through Fourier domain multiplication, and incorporating rotation-based channelwise self-attention mechanisms. This method enhances the model's ability to discern complex interdependencies between source images.

TABLE I

COMPARISON ON THE WORDVIEW-II AND WORDVIEW-III DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ↑ INDICATES THAT THE LARGER THE VALUE, THE BETTER THE PERFORMANCE, AND ↓ INDICATES THAT THE SMALLER THE VALUE, THE BETTER THE PERFORMANCE

Method	WordView-II						WordView-III					
	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	SCC ↑	Q ↑	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	SCC ↑	Q ↑
SFIM	34.1297	0.8975	0.0439	2.3449	0.9079	0.6064	21.8212	0.5457	0.1208	8.9730	0.6952	0.4531
GS	35.6376	0.9176	0.0423	1.8774	0.9225	0.6307	22.5608	0.5470	0.1217	8.2433	0.7131	0.4411
Brovey	35.8646	0.9216	0.0403	1.8238	0.8913	0.6163	22.5060	0.5466	0.1159	8.2331	0.7033	0.4394
IHS	35.2962	0.9027	0.0461	2.0278	0.8534	0.5704	22.5579	0.5354	0.1266	8.3616	0.6994	0.4301
GFPCA	34.558	0.9038	0.0488	2.1400	0.8924	0.4665	22.3400	0.4826	0.1294	8.3964	0.6987	0.3115
PNN	40.755	0.9624	0.0259	1.0646	0.9677	0.7426	29.9418	0.9121	0.0824	3.3206	0.9540	0.8679
PANNet	40.8176	0.9626	0.0257	1.0557	0.9680	0.7437	29.6840	0.9072	0.0851	3.4263	0.9512	0.8631
MSDCNN	41.3355	0.9664	0.0242	0.9940	0.9721	0.7577	30.3038	0.9184	0.0782	3.1884	0.9577	0.8763
SRPPNN	41.4538	0.9679	0.0233	0.9899	0.9729	0.7691	30.4346	0.9202	0.0770	3.1553	0.9581	0.8776
GPPNN	41.1622	0.9684	0.0244	1.0315	0.9722	0.7627	30.1785	0.9175	0.0776	3.2593	0.9569	0.8739
INNformer	41.6903	0.9704	0.0227	0.9514	0.9753	0.7756	30.5365	0.9225	0.0747	3.0997	0.9569	0.8817
MutNet	41.6773	0.9705	0.0224	0.9519	0.9724	0.7741	30.4907	0.9223	0.0749	3.1125	0.9569	0.8802
SFINet	41.7244	0.9725	0.0220	0.9506	0.9720	0.7751	30.5971	0.9236	0.0741	3.0798	0.9569	0.8802
PanFlowNet	41.8548	0.9712	0.0224	0.9335	0.9735	0.7767	30.4873	0.9221	0.0751	3.1142	0.9601	0.8815
BugPan	41.7539	0.9685	0.0231	0.9475	0.9731	0.7627	30.6123	0.9206	0.0759	3.0708	0.9603	0.8768
ADWM	41.8919	0.9711	0.0221	0.9284	0.9749	0.7781	30.6274	0.9228	0.0738	3.0654	0.9605	0.8821
Ours	**42.1357**	**0.9726**	**0.0212**	**0.9047**	**0.9775**	**0.7874**	**30.7159**	**0.9235**	**0.0734**	**3.0460**	**0.9611**	**0.8828**

E. Omni-Attention High-Order Module

Building upon the previous analysis, we present a theoretically viable omni-attention high-order module, as depicted in Fig. 7. The proposed high-order block computes score matrices for both spatial and channel directions through sequential matrix operations and rotations. Given the PAN-modality feature $\mathbf{F}_P \in \mathbb{R}^{C\times HW}$ and the MS-modality feature $\mathbf{F}_{\text{ms}} \in \mathbb{R}^{HW\times C}$, the module calculates omni-attention across spatial and channelwise interactions. Initially, we apply the equivalently efficient form in the spatial domain to achieve this

$$\mathbf{A}_s = \mathcal{F}^{-1}\left(\mathcal{F}\left(F_{\text{ms}}\right) \odot \mathcal{F}\left(F_P\right)\right) \tag{51}$$

$$\text{Norm}\left(\mathbf{A}_s\right) = \text{LN}\left(\mathbf{A}_s\right). \tag{52}$$

Simultaneously, we rotate the input query and key matrices to obtain their transposed forms. We also rotate the value matrices, resulting in the value matrix $\mathbf{V} \in \mathbb{R}^{C\times HW}$ for subsequent channelwise self-attention. The resulting channelwise attention map, of size $\mathbb{R}^{C\times C}$, effectively models the relationships across channels

$$\mathbf{A}_c = \text{softmax}\left(\frac{\mathbf{F}_{\text{ms}} \otimes \mathbf{F}_P^T}{\sqrt{d_k}}\right). \tag{53}$$

Next, we obtain the first-order aggregated result $\mathbf{V}_1$ by the inverse rotation of the channel attention output

$$\mathbf{V}_{1b} = \mathbf{A}_c^1 \otimes \mathbf{F}_P, \quad \mathbf{V}_1 = \mathbf{V}_{12}^T \odot \mathbf{A}_s^1 \tag{54}$$

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TABLE II
QUANTITATIVE COMPARISONS ON THE TNO DATASET

Metrics	DDcGAN	DenseFuse	AUIF	DIDFuse	ReCoNet	SDNet	TarDAL	U2Fusion	UMFusion	LRRNet	TIM	CDDFuse	SwinFuse	SuperFuse	HOAB
SF	0.0476	0.0374	0.0508	0.0489	0.0293	0.0496	0.0511	0.0355	0.0358	0.0352	0.0368	0.0547	0.0458	0.0386	**0.0563**
MI	1.8470	2.4018	2.2714	2.4422	2.4263	2.1860	2.6480	1.9225	2.2474	2.5288	3.6644	2.9519	3.2072	2.9762	**3.6859**
VIF	0.6737	0.7997	0.8146	0.8286	0.8266	0.7624	0.8601	0.6878	0.7169	0.7719	0.8078	0.9083	0.8983	0.7996	**0.9264**
AG	4.8312	3.6337	4.7120	4.6358	3.3227	4.8438	4.8871	3.7329	3.1660	3.5301	3.3577	5.0782	4.5107	3.2637	**5.3047**
Qabf	0.3492	0.4227	0.4256	0.4046	0.3570	0.4210	0.4671	0.4178	0.3984	0.3382	0.4342	0.5022	0.5007	0.4352	**0.5056**
FMI	0.858	0.870	0.879	0.863	0.878	0.883	0.881	0.879	0.888	0.8788	0.892	0.8855	0.8842	0.884	**0.9026**

TABLE III
QUANTITATIVE COMPARISONS ON THE M3FD DATASET

Metrics	DDcGAN	DenseFuse	AUIF	DIDFuse	ReCoNet	SDNet	TarDAL	U2Fusion	UMFusion	LLRNet	TIM	CDDFuse	SwinFuse	SuperFuse	HOAB
SF	0.0613	0.0368	0.0464	0.0551	0.0413	0.0533	0.0491	0.0420	0.0344	0.0380	0.0442	0.0579	0.0536	0.0405	**0.0631**
MI	2.5397	2.9297	3.0490	3.0476	3.0495	3.2314	3.1624	2.7590	3.0871	2.8813	3.5091	3.9090	4.4006	3.4421	**4.5132**
VIF	0.7684	0.7621	0.8192	0.8770	0.8184	0.6784	0.8100	0.7091	0.7089	0.7148	0.8624	0.8946	0.8783	0.7532	**0.9123**
AG	5.2750	3.3028	3.9219	4.8714	3.9289	4.7299	4.1356	3.9673	2.9283	3.2075	3.7289	4.8739	4.6014	3.4993	**5.3940**
Qabf	0.4804	0.5031	0.4798	0.4941	0.4858	0.5283	0.4052	0.5378	0.3966	0.5057	0.5376	0.5282	0.5426	0.4771	**0.5412**
FMI	0.836	0.863	0.845	0.831	0.845	0.846	0.825	0.850	0.855	0.8870	0.8852	0.8749	0.8840	0.8520	**0.900**

$$\mathbf{A}_c^2 = \texttt{Conv}_{1\times3}\left(\mathbf{A}_c^1\right),\ \ \mathbf{A}_s^2 = \text{LN}\left(\texttt{Conv}_{1\times1}\left(\mathbf{A}_s^1\right)\right) \tag{55}$$

$$\mathbf{V}_{2b} = \mathbf{V}_1 \odot \mathbf{A}_s^2,\ \ \mathbf{V}_1 = \mathbf{A}_s^2 \otimes \mathbf{V}_{2b}^T. \tag{56}$$

Building on the aforementioned attention-sharing strategy, we achieve high-order spatial and channelwise attention modeling as follows:

$$\mathbf{A}_c^i = \texttt{Conv}_{1\times3}\left(\mathbf{A}_c^{i-1}\right),\ \ \mathbf{A}_s^i = \text{LN}\left(\texttt{Conv}_{1\times1}\left(\mathbf{A}_s^{i-1}\right)\right) \tag{57}$$

$$\mathbf{V}_{ib} = \mathbf{V}_1 \odot \mathbf{A}_s^i,\ \ \mathbf{V}_{i+1} = \mathbf{A}_s^i \otimes \mathbf{V}_{ib}^T \tag{58}$$

where $\mathcal{T}(\cdot)$ denotes the rotation operation. This design exhibits compelling properties by integrating elementwise results of two matrix operations, facilitating omni-axial interactions.

To facilitate cross-order information integration, we enhance the representation of cross-modality interactions by incorporating diverse information in a cross-order manner, as illustrated in Fig. 7. This enhancement enables the generation of more informative representations by leveraging the observation that different orders tend to capture diverse and complementary patterns. This approach ensures a richer and more comprehensive modeling of the interaction between modalities, ultimately leading to improved performance

$$\mathbf{V}_b \leftarrow \texttt{Conv}\left(\text{concat}\left[\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_n\right]\right). \tag{59}$$

Finally, pan-sharpening aims to super-resolve the low-resolution MS image using the texture-rich PAN image, making the MS image the prominent modality. To retain essential information, we implemented residual information memorizing mechanism within the prominent modality, ensuring critical details are preserved throughout the network's learning process

$$F_{\text{em}} \leftarrow \mathbf{V}_b + F_{\text{ms}}. \tag{60}$$

IV. EXPERIMENTS

A. Experimental Settings

To assess the effectiveness of our HOAB method, we conducted comprehensive experiments on three publicly M3FD [48], RoadScene [46], and TNO [49]. We compare our method with state of the arts, including DDcGAN [43], DenseFuse [40], AUIF [50], DIDFuse [51], ReCoNet [52], SDNet [53], TarDAL [48], U2Fusion [46], UMFusion [54], LRRNet [55], CDDFuse [56], SwinFuse [6], SuperFusion [57], and TIM [47] across three datasets. We implemented our methods using PyTorch on a single NVIDIA GTX 3090 GPU. The Adam optimizer, with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.99$, was utilized to update the model over 30 000 iterations, each with a batch size of 8. The initial learning rate was set to 1×10^{-4} and decreased by a factor of 0.5 every 5000 iterations. The training patch size was configured to 128×128 pixels. A high-quality fused image should capture both salient objects and visual quality from the multimodal images. To comprehensively measure the fusion results, we employed six metrics: spatial frequency (SF), MI, visual information fidelity (VIF), average gradient (AG), Q_{abf}, and feature MI (FMI). Higher values for these metrics indicate superior fusion performance.

To rigorously assess the performance of the HOAB, we undertake an extensive comparative analysis against a range of benchmark pan-sharpening methods. The traditional approaches in our evaluation encompass SFIM [58], Brovey [59], GS [60], IHS [61], and GFPCA [62]. Additionally, we include various DL-based techniques, such as PNN [19], PANNET [20], MSDCNN [22], SRPPNN [23], GPPNN [24], MutNet [28], INNformer [7], BugPan [63], ADWM [64], SFINet [65], and PanFlowNet [66]. During training, we employed the Adam optimizer for 1000 epochs with a batch size of 4 and an initial learning rate of 5×10^{-4}, reducing it by half every 200 epochs

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TABLE IV
QUANTITATIVE COMPARISONS ON THE ROADSCENE DATASET

Metrics	DDcGAN	DenseFuse	AUIF	DIDFuse	ReCoNet	SDNet	TarDAL	U2Fusion	UMFusion	LRRNet	TIM	CDDFuse	SwinFuse	SuperFuse	HOAB
SF	0.0393	0.0403	0.0549	0.0515	0.0327	0.0545	0.0502	0.0471	0.0377	0.0378	0.0369	0.0620	0.0433	0.0447	**0.0661**
MI	2.6178	3.1276	3.1110	3.1840	3.1594	3.4225	3.4640	2.8110	3.2019	2.7687	3.6291	3.2083	3.5534	3.4463	**3.5933**
VIF	0.5946	0.8025	0.8466	0.8274	0.7956	0.8207	0.7872	0.7402	0.7913	0.7744	0.8246	0.8792	0.8356	0.8279	**0.8868**
AG	3.7710	4.0123	5.3179	5.0377	3.4841	5.5307	4.3890	4.8851	3.7154	3.6054	3.3687	5.3786	4.0679	4.1670	**6.1740**
Qabf	0.3093	0.5133	0.5166	0.4870	0.3943	0.5329	0.4484	0.5258	0.5055	0.3317	0.3780	0.4976	0.4712	0.5132	**0.5263**
FMI	0.859	0.868	0.856	0.853	0.858	0.863	0.852	0.861	0.866	0.8611	0.852	0.8610	0.8682	0.865	**0.8746**

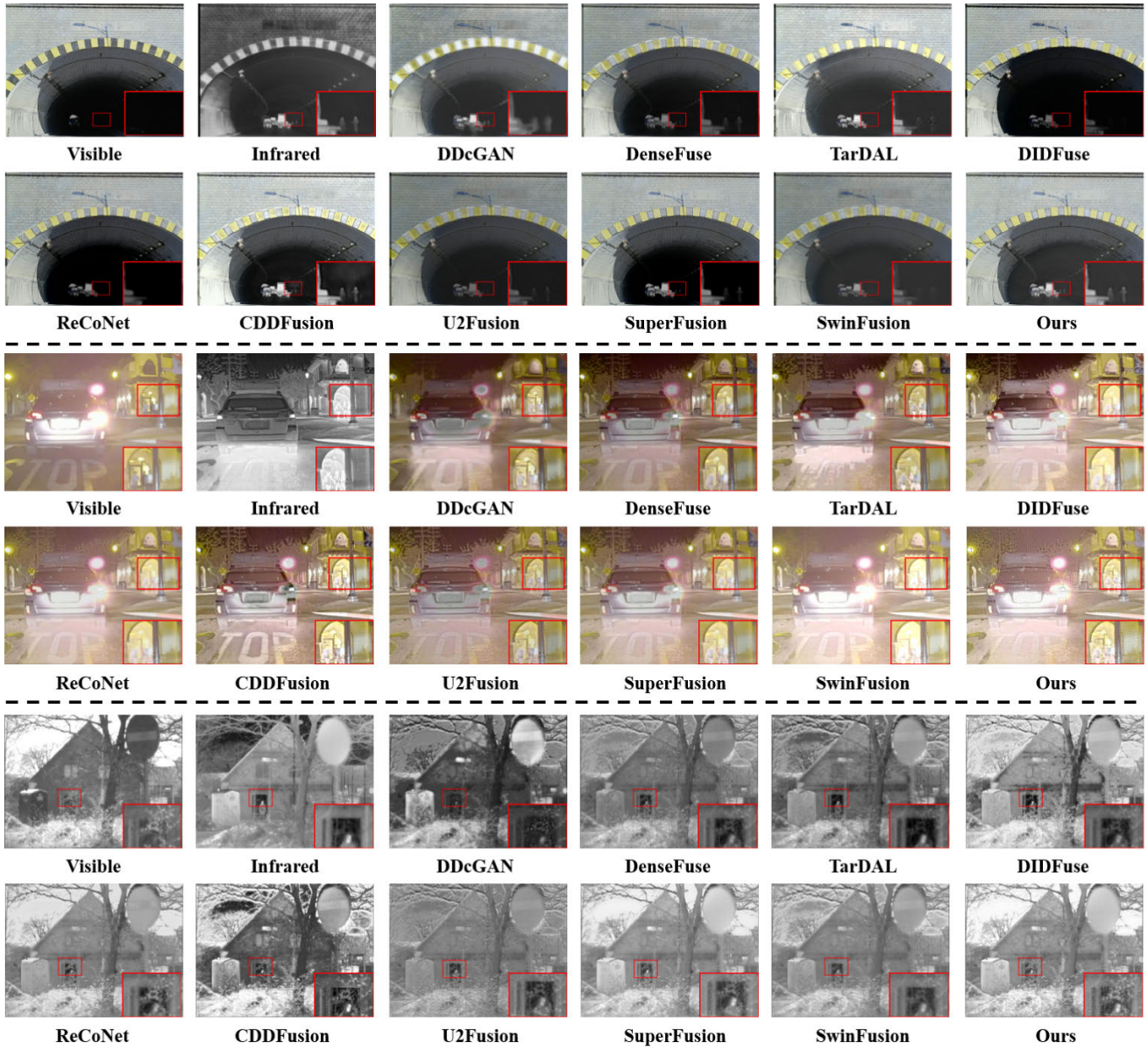


Fig. 8. Qualitative results of different fusion methods on M3FD, RoadScene, and TNO datasets, respectively.

to ensure stable convergence. Due to the lack of ground-truth pan-sharpened images, we used the Wald protocol tool [67] to generate the training dataset. Our evaluation was conducted on the Worldview-II and Worldview-III datasets, where PAN images were cropped into 128×128 pixel patches and the corresponding MS patches were resized to 32×32 pixels. To evaluate performance, we utilized a comprehensive set of image quality assessment (IQA) metrics, including the relative dimensionless global error in synthesis (ERGAS) [68], peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), correlation coefficient (SCC), Q index, spectral angle mapper (SAM) [69], spectral distortion index (D_λ), spatial distortion index (D_S), and quality without reference (QNR).

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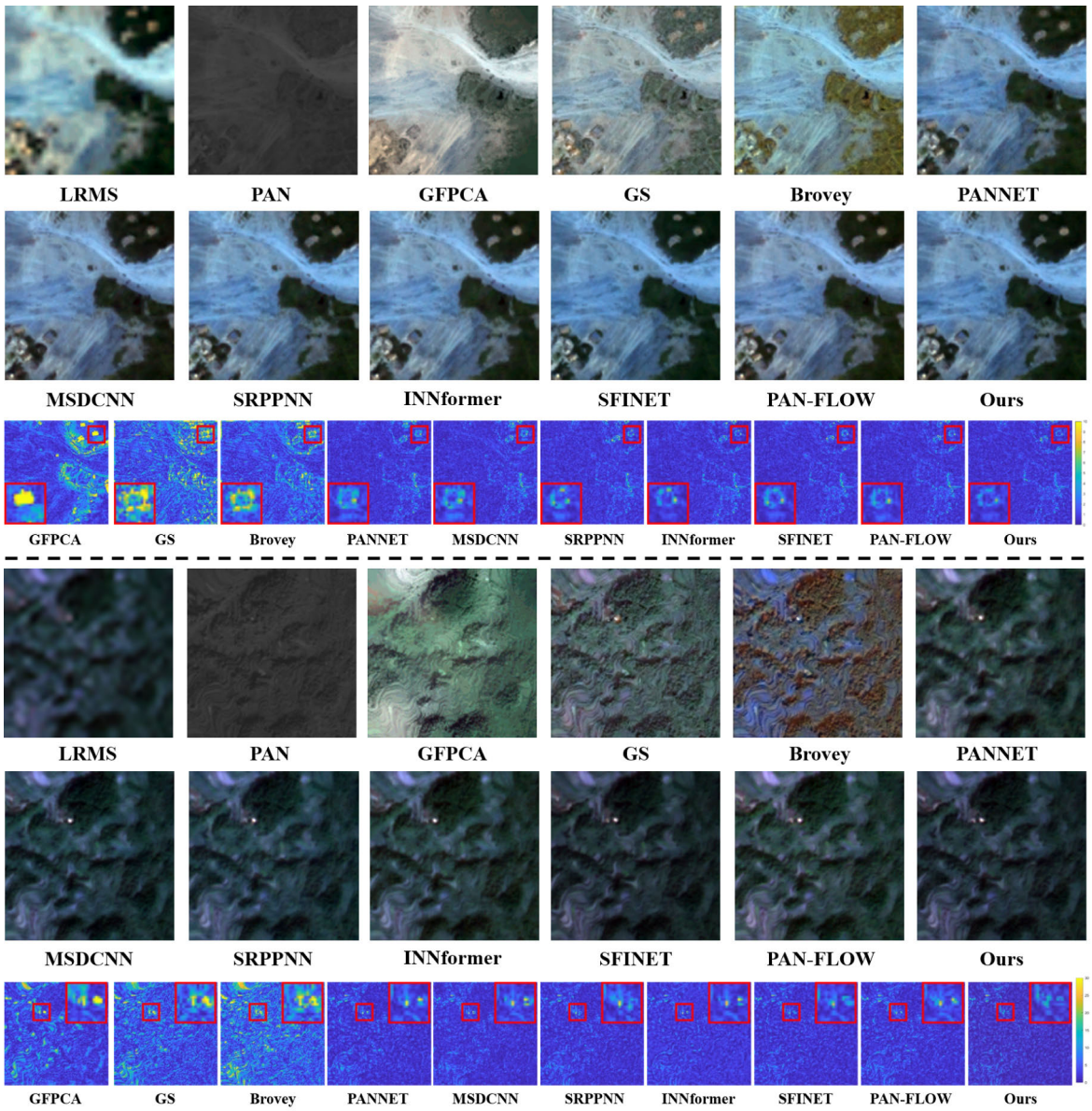


Fig. 9. Visual comparisons between other pan-sharpening methods on WorldView-II and WorldView-III satellite.

TABLE V
RESULTS OF OBJECT DETECTION ON THE M3FD DATASET

Methods	Lamp	Car	Bus	Motor	Truck	People	mAP
ReCoNet	0.336	0.744	0.482	0.311	0.445	0.556	0.479
SDNet	0.265	0.702	0.429	0.344	0.331	0.515	0.431
U2Fusion	0.312	0.724	0.475	0.352	0.392	0.534	0.465
UMFusion	0.254	0.692	0.424	0.309	0.343	0.518	0.423
HOAB	**0.344**	**0.742**	**0.502**	**0.357**	**0.419**	**0.543**	**0.484**

B. Comparisons With State of the Arts

This section details the evaluation of our proposed pan-sharpening techniques, conducted on two prominent satellite datasets: WorldView-II and WorldView-III. To thoroughly evaluate the performance of our methods, we employed a comprehensive set of metrics, and the results are systematically presented in Table I. These tables underscore the exceptional performance of our methods, clearly illustrating their superiority over the benchmark algorithms across all considered evaluation criteria. Specifically, our proposed methods exhibit peak PSNR improvements of 0.28 and 0.23 dB compared to the highest-performing algorithm for the WorldView-II and WorldView-III datasets, respectively. This enhancement in PSNR reflects a significant advancement in image quality, demonstrating the efficacy of our approach in achieving superior detail preservation and clarity. Moreover, our methods achieve notable improvements across additional evaluation metrics, reinforcing their superiority over the existing state-of-the-art DL-based approaches. The consistency of these enhancements across different metrics further validates the robustness and efficacy of our methods.

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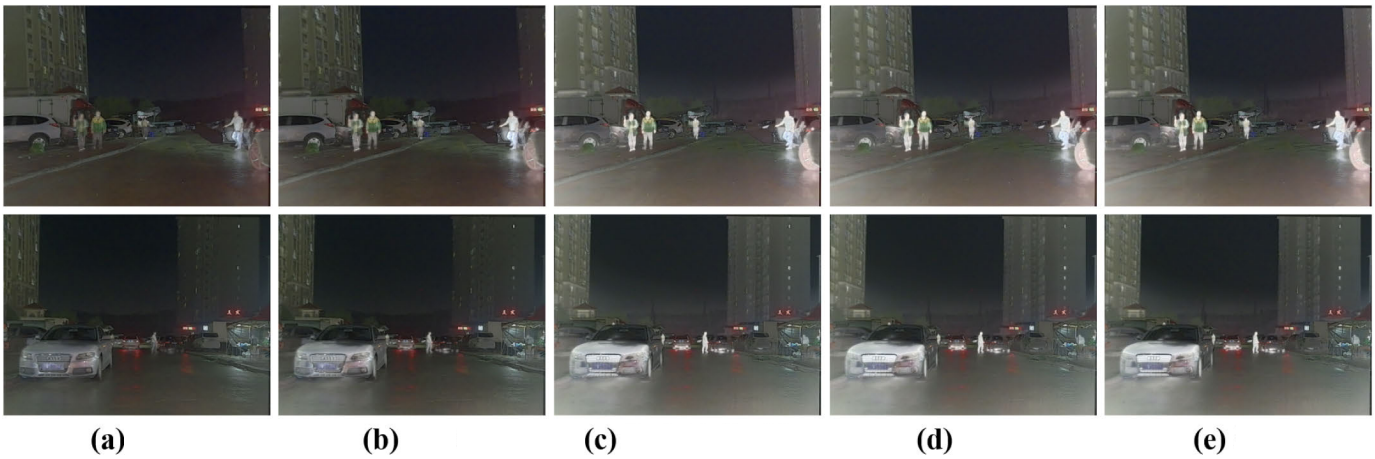


Fig. 10. Qualitative results of ablation studies about the number of iterations L. With fewer iterations, the outline of salient targets is blurry; however, as the iterations increase, contour highlights become more accurate. (a) $L = 1$. (b) $L = 2$. (c) $L = 3$ (default). (d) $L = 4$. (e) $L = 5$.

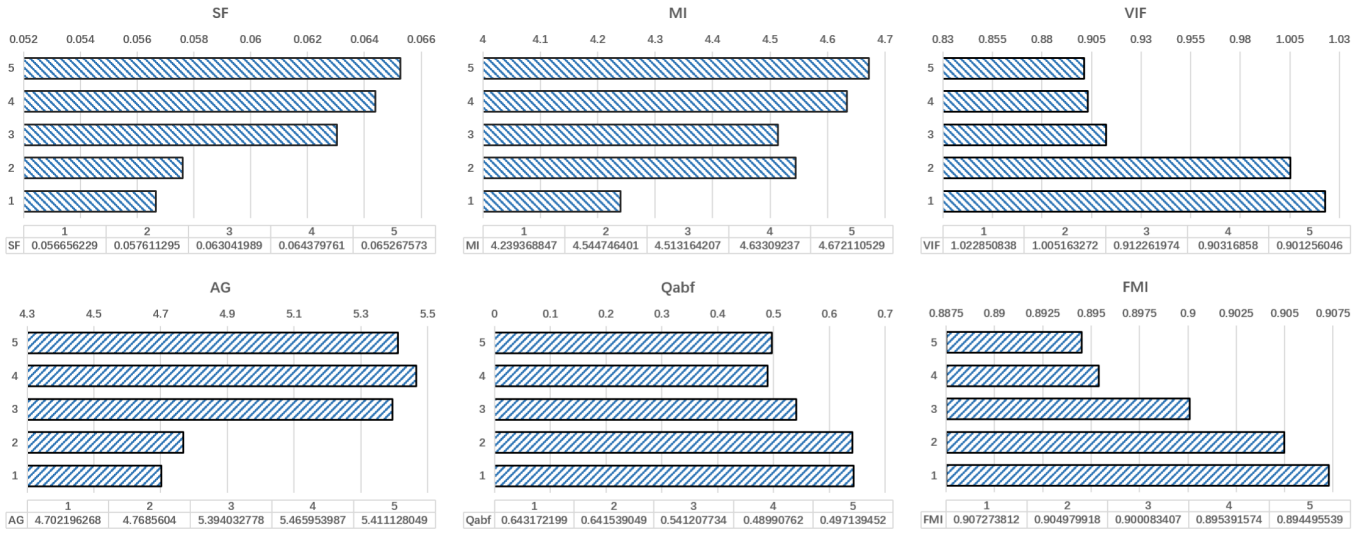


Fig. 11. Ablation studies of the proposed SHIP with different numbers of order N on the M3FD dataset.

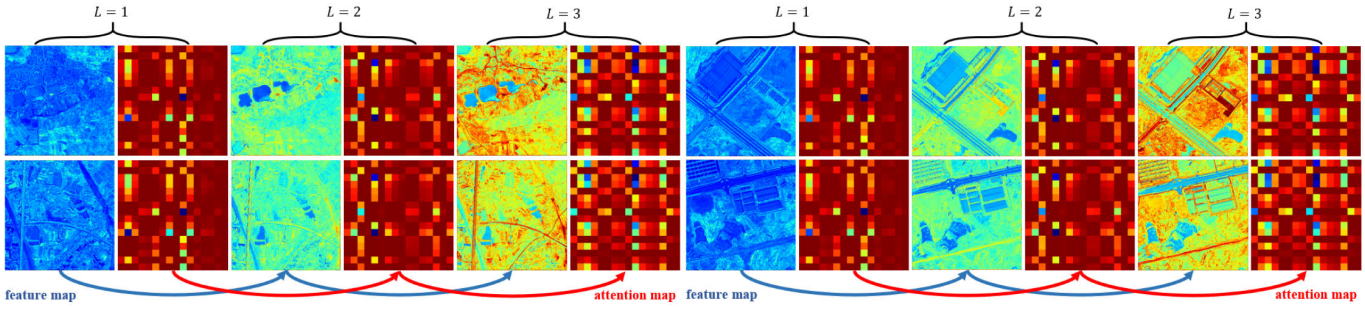


Fig. 12. Feature visualization and attention map within each HOAB. It illustrates the efficacy of high-order interactions in two perspectives: 1) as the number of iterations **L** increases, feature responses intensify, emphasizing salient objects and enhancing informative representation; and 2) the corresponding attention maps exhibit clear diversity in feature representations and information exploration, confirming that our proposed strategy effectively mitigates attention collapse.

To further validate the effectiveness of our approach, we provide visual comparisons of representative samples from the WorldView-II and WorldView-III datasets, as illustrated in Fig. 9. The final row of each figure presents the mean squared error (mse) residues between the pan-sharpened images and the ground truth. These visual comparisons demonstrate that our method substantially reduces both spatial and spectral distortions when compared to alternative techniques, as evidenced by the mse residue maps. The mse residues indicate that our method closely approximates the ground truth with a higher

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TABLE VI
COMPARISON OF VARIOUS IMAGE FUSION METHODS BASED ON DIFFERENT METRICS

Metrics	DDcGAN	DenseFuse	DIDFuse	AUIF	SDNet	TarDAL	U2Fusion	UMFusion	TIM	LRRNet	CDDFuse	ReCoNet	SuperFusion	SwinFusion	Ours
Par. (M)	1.097	0.074	0.012	0.373	0.067	0.297	0.659	0.629	1.232	0.049	1.783	0.209	0.139	9.275	0.132
FLOPs (G)	896.840	48.960	0.014	103.560	37.350	82.370	366.340	174.690	194.630	12.809	495.248	12.540	11.920	276.705	17.761
Time (s)	0.211	0.251	0.166	0.118	0.045	0.001	0.123	0.044	0.145	0.020	0.483	0.051	0.037	2.131	0.038

degree of fidelity than other methods. Notably, the zoomed-in regions of the mse maps reveal that our method tends to exhibit a bluish hue, which contrasts with the more yellowish hue seen in outputs from competing methods, suggesting superior spatial fidelity and reduced distortion. Collectively, these visual assessments provide compelling evidence that our proposed method not only achieves the state-of-the-art performance but also surpasses the existing pan-sharpening algorithms in both quantitative and qualitative evaluations.

Our HOAB method demonstrates exceptional quantitative performance across various metrics, as shown in Tables II, III, and IV. It achieves the highest scores in MI and FMI, reflecting its superior ability to integrate and transfer information from both infrared and visible source images into the fused output, resulting in a richer and more informative fusion. Furthermore, our method leads in SF, AG, and Q_{abf} metrics, indicating its effectiveness in preserving intricate texture details and enhancing overall image clarity. The high scores in these metrics emphasize our approach's robustness in combining complementary information from different modalities. The VIF scores further confirm the high visual quality and minimal distortion of our fusion results, making them visually appealing and practically useful. Notably, our HOAB method surpasses SDNet in SF by a significant margin, highlighting its ability to capture and maintain fine-grained details in complex scenarios. The consistently high performance across these evaluation metrics underscores the robustness and effectiveness of our HOAB method in diverse settings. To further assess the effectiveness of our approach, we utilized FCOS as the detection model to evaluate object detection performance on the fused results. The results, presented in Table V, demonstrate the superior performance in terms of mAP, highlighting its capability to enhance detection outcomes through improved image fusion.

Fig. 8 presents the qualitative comparisons of three typical image pairs from various datasets, illustrating the superior performance of our HOAB method. Our approach excels in retaining and highlighting prominent features from both infrared and visible images, resulting in fusion outputs with high contrast and clearly defined contours. For instance, our method enhances the visibility of cars and tree branches, providing a high level of detail that is essential for accurate visual interpretation. Compared to other methods, such as DenseFuse, SDNet, and U2Fusion, which struggle to emphasize discriminative targets, our results consistently exhibit better contrast and clearer texture details. DDcGAN and ReCoNet, while effective in some aspects, often fail to capture the fine textural information that our method successfully preserves. The qualitative results show that our HOAB method produces images with brighter, more detailed features, particularly in the foreground objects framed by red and green boxes. Unlike DenseFuse and DIDFuse, which capture thermal features but lack brightness, our method ensures that the fusion outputs are both informative and visually appealing. Overall, our HOAB method achieves an optimal balance between highlighting salient features and preserving detailed textures.

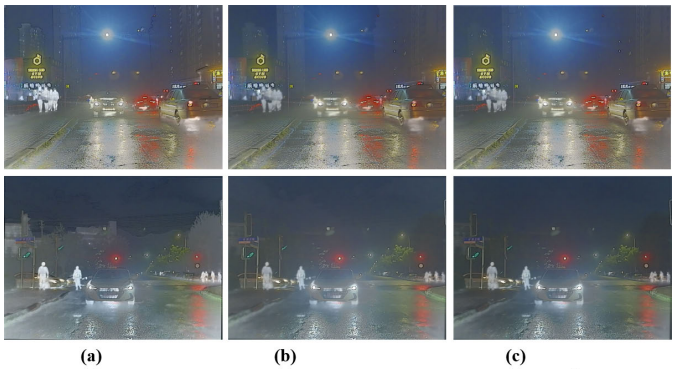


Fig. 13. Ablation studies of the proposed SHIP with different numbers of block L on the M3FD dataset. (a) Ours, (b) without channel, and (c) without spatial.

C. Ablation Studies

1) Effect of the Number of Iterations L: We conduct the experiment to investigate the effect of the number of blocks on the results from varying $\mathbf{L} = \mathbf{1}, \ldots, \mathbf{5}$. From the observations in Table VI, it is evident that the model's performance improved considerably with an increase in the number of blocks. However, further increments in L resulted in a decreasing trend in Q_{abf} and AG, possibly due to challenges in gradient propagation. Consequently, we employ $L = 3$ as the default block number in all experiments. In addition, we illustrate the qualitative visualizations in Fig. 10. The results indicate that as the number increases, the prominence of key objects is enhanced, leading to improved overall visual perception. This is evident in Fig. 12, where the feature maps become increasingly informative, and the diversity of the attention maps enriches with the growing number.

2) Ablation Studies About the Spatial and Channel High-Order Interactions: We independently examine the necessity of spatial fine-grained and channelwise high-order 1 synergy interactions. Table VII illustrates a significant decrease in performance when either spatial or channel dimensions are removed. Consequently, the collaborative involvement of both spatial and channel dimensions is essential for effectively aggregating spatial fine-grained correlations and distinguishing channelwise interdependencies between different modalities.

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TABLE VII
ABLATION STUDIES ABOUT THE SPATIAL AND CHANNEL HIGH-ORDER INTERACTIONS

Config	SF↑	MI↑	VIF↑	AG↑	Q_{abf}↑	FMI↑
w/o Spatial	0.0568	4.3659	0.9309	4.8155	0.6242	0.8986
w/o Channel	0.0574	3.9532	0.9564	4.7760	0.6227	0.9007

In Fig. 13, the visualization indicates that deleting the spatial or channel dimensions will result in brightness degradation and weaken the prominence of salient objects, thus verifying its necessity.

V. CONCLUSION

In this article, we introduce the high-order omni-attention paradigm for multimodal image fusion, addressing the limitations of current methods focused on second-order, uni-dimensional interactions. Our approach enhances fusion by introducing two key innovations: a high-order interaction mechanism that reduces computational load by calculating attention scores once and sharing them across layers and omni-dimensional integration, where each pixel is treated as a token in 3-D space. This allows for dense interactions in the Fourier domain and channelwise self-attention, capturing complex interdependencies between source images. Extensive experiments on benchmarks demonstrate that HOAB consistently outperforms the state-of-the-art methods, proving the effectiveness of combining high-order interactions with omni-dimensional attention for superior fusion performance.

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